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Chemical Bonding & Hybridisation — D4 Issue #38

Candidate Specimen: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In formamide, $\text{H}-\text{C}(=\text{O})-\text{NH}_2$, a local Lewis count gives nitrogen three sigma bonds

Representation

and a lone pair. You are also given that the approximately planar amide arrangement permits nitrogen-lone-pair donation into the adjacent carbonyl pi system, which requires a suitably aligned p-like orbital. Choose a model that accounts for both the sigma framework and delocalisation. Resolve the apparent conflict with an automatic 'four domains therefore sp³' label and state the limits of your chosen hybridisation description.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Supplied formamide connectivity



Formamide connectivity; : is

Connectivity layout only; page angles are not measured g

Supplied situation

Need a hint? • guided support

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Common wrong route · open if you want a warning

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Need the concept again?

Delocalisation-sensitive hybridisation boundaries

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the formamide situation supplied in Q1, construct one qualitative representation connecting the Lewis lone pair on nitrogen, the nitrogen orbital used for donation and

Representation

the carbonyl pi system. Then construct the corresponding approximately 90-degree-twisted arrangement about the C–N bond. Preserve atom connectivity and distinguish changes in orbital alignment from changes in formal electron count. Do not claim these sketches alone measure the donation strength.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Fixed connectivity for two arrangements



Formamide connectivity; : is

Construct reference and twisted arrangement:

Connectivity layout only; page angles are not measured g

Supplied situation

Need a hint? · guided support

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Need the concept again?

Conformational torsion and p-orbital alignment

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies a qualitative model result: with atom connectivity fixed, twisting a planar amide arrangement toward 90 degrees reduces the nitrogen-to-carbonyl orbital alignment and the delocalisation contribution. Explain that

Representation

result through orbital overlap. Assess whether 'the hybridisation label changes, so donation disappears' is a sufficient explanation, and identify which parts of your account are model assumptions rather than measured facts.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Supplied qualitative twist result

Fixed atom connectivity

Given: twist toward 90° reduces alignment

Given: delocalisation contribution decreases

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Conformational torsion and p-orbital alignment

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Compare two nitrogen sites: A is the amide nitrogen in $\text{H}-\text{C}(=\text{O})-\text{NH}_2$; B is the amine nitrogen in $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$. In the simplified introductory model, the intervening saturated CH_2 group has no continuous aligned p-orbital pathway.

Representation

Develop a decision procedure combining local sigma/lone-pair counting, adjacency, orbital alignment and the model's validity conditions. Apply it to A and B, then state a counterfactual structural change that would require you to reassess your procedure. Do not equate spatial proximity with conjugation.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Supplied direct/spacer comparison



Formamide connectivity; : is the N lone pair



Saturated spacer; no cha

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? · guided support

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Need the concept again?

Local domain counting versus conjugated networks

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student claims: 'Hybridisation uniquely determines bond angles and any deviation proves the hybridisation label wrong.' The benchmark supplies approximate observations: CH₄ tetrahedral about 109.5 degrees; NH₃ pyramidal about 107 degrees; H₂O bent about 104.5 degrees; all three commonly use an introductory sp³

Representation

description at the central atom. It also supplies approximately planar amide nitrogen with lone-pair delocalisation. Construct and test a qualified replacement claim using these observations, local domain counting, lone-pair effects and the amide boundary case. State what further evidence would be needed before treating a qualitative orbital picture as a unique physical account.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Supplied counterexample observations

CH₄: ~109.5°; NH₃: ~107°; H₂O: ~104.5°

Given: common introductory sp³ description

Given: approximately planar amide N with lone pair

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? • guided support

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Common wrong route · open if you want a warning

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Need the concept again?

Delocalisation-sensitive hybridisation boundaries

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the two nitrogen sites A and B in Q4. State the conditions under which the local four-domain counting rule is a useful introductory classification, and the evidence that would make its automatic

Representation

use inadequate. Distinguish recalling a rule from deciding whether its conditions hold.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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The two supplied nitrogen sites



Formamide connectivity; : is the N lone pair



Saturated spacer; no cha

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? · guided support

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Need the concept again?

Local domain counting versus conjugated networks

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A neutral methyl radical, $\text{CH}_3\cdot$, is supplied as approximately planar, with the unpaired electron in a p-like orbital perpendicular to the three C–H sigma directions. Translate that supplied model into an orbital sketch and assess whether 'three sigma bonds always leave an empty p orbital' follows. Keep the given model distinct from a universal claim.

Representation

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route • open if you want a warning

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Supplied methyl-radical model

CH₃• is supplied as approximately planar

Given: one unpaired electron in a perpendicular

Three C-H sigma directions

Connectivity layout only; page angles are not measured g
Supplied situation

Need a hint? • guided support

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Need the concept again?

Delocalisation-sensitive hybridisation boundaries

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Two qualitative descriptions of formamide are proposed: a completely localised pyramidal nitrogen lone pair, and an approximately planar nitrogen arrangement with possible lone-pair donation into the carbonyl pi system.

Representation

Propose observations that could discriminate between them, explain the warranted inferences, and identify at least one inference that geometry alone cannot establish.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route • open if you want a warning



Two descriptions to discriminate

A: localised pyramidal nitrogen lone pair

B: approximately planar N; possible carbor

Propose observations; no measured strength

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? • guided support



Need the concept again?

Delocalisation-sensitive hybridisation boundaries

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies a simplified torsional model in which the relevant p-orbital overlap factor varies as $\cos(\theta)$, where θ is the angle between their alignment directions. Compare $\theta = 0$, 60 and 90 degrees, then assess the claim

Representation

that the molecule's total energy must therefore vary as $\cos(\theta)$. Separate execution of the given model from a justified claim about the physical system.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Supplied cosine-factor model

Given: $g = \cos(\theta)$

Compare $\theta = 0^\circ, 60^\circ, 90^\circ$

Assess a separate claim about total mole

Connectivity layout only; page angles are not measu

Supplied situation

Need a hint? · guided support

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Need the concept again?

Conformational torsion and p-orbital alignment

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A proposed teaching rule says, 'Count sigma bonds and lone pairs, assign a hybridisation label, and the label explains every shape and reactivity difference.'
Audit this rule against NH₃, allene, formamide and the separated amine site

Representation

H–C(=O)–CH₂–NH₂. Produce a qualified decision framework, identify where local counting is useful, and state what evidence or model is needed beyond it.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Supplied universal-rule audit cases

NH₃; allene; formamide; saturated-space

Proposed rule: count → label → every sh

Audit the rule and construct a qualified pr

Connectivity layout only; page angles are not measu

Supplied situation

Need a hint? · guided support

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Need the concept again?

Local domain counting versus conjugated networks

Chemistry · Chemical Bonding & Hybridisation — D4 Issue #38 Candidate Specimen